

Study Guide - SL - Topic B5 D2 D3 D4 - Paper 2 [125 marks]

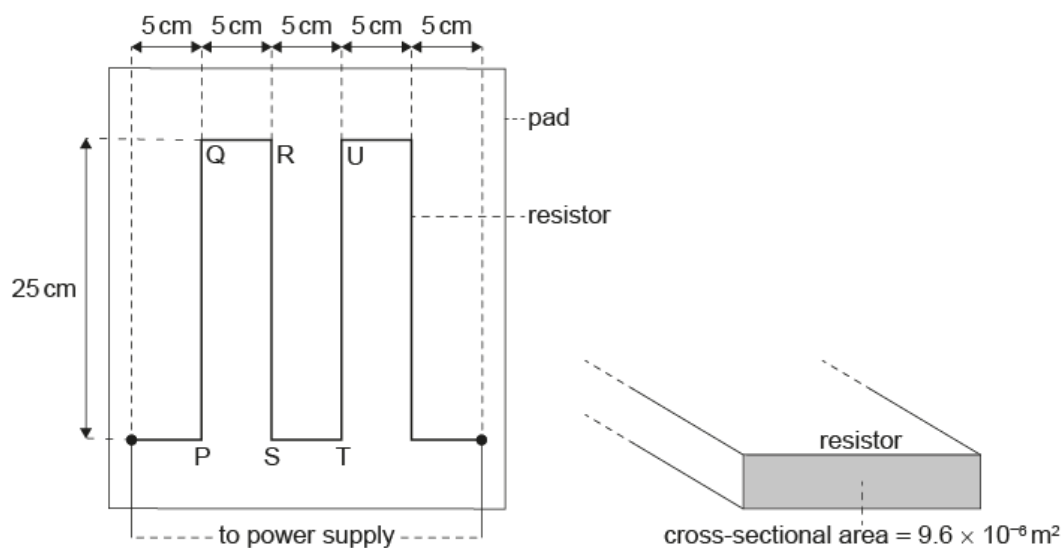
1. [Maximum mark: 12]

23M.2.SL.TZ2.4

An electrically heated pad is designed to keep a pet warm.

The pad is heated using a resistor that is placed inside the pad. The dimensions of the resistor are shown on the diagram. The resistor has a resistance of $4.2\ \Omega$ and a total length of $1.25\ \text{m}$.

diagram not to scale



When there is a current in the resistor, the temperature in the pad changes from a room temperature of $20\ ^\circ\text{C}$ to its operating temperature at $35\ ^\circ\text{C}$.

- (a) The designers state that the energy transferred by the resistor every second is $15\ \text{J}$.

Calculate the current in the resistor.

[1]

Markscheme

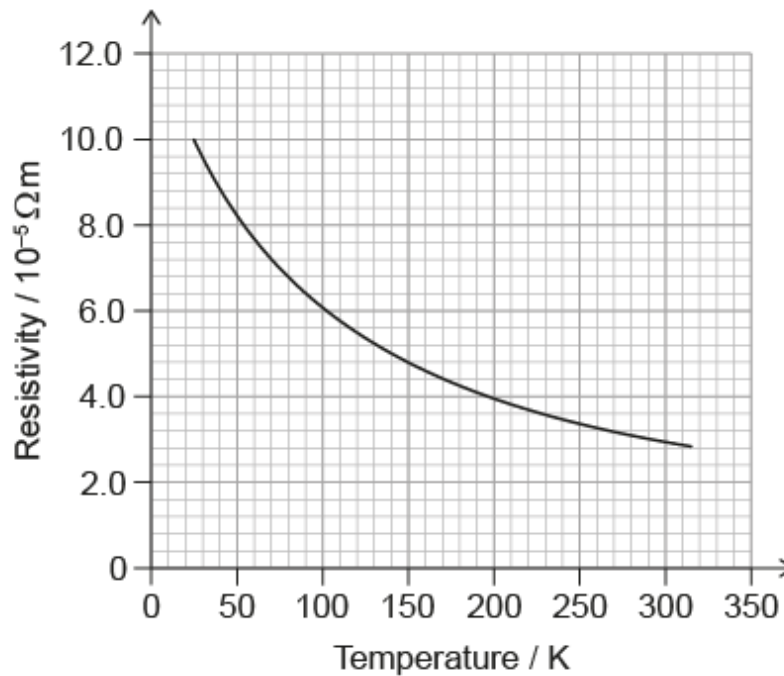
$$I = \sqrt{\frac{P}{R}} = 1.9\ \text{A} \checkmark$$

Examiners report

This was well answered.

- (b) The designers wish to make the resistor from carbon fibre.

The graph shows the variation with temperature, in Kelvin, of the resistivity of carbon fibre.



- (b.i) The resistor has a cross-sectional area of $9.6 \times 10^{-6} \text{ m}^2$.

Show that a resistor made from carbon fibre will be suitable for the pad.

[3]

Markscheme

ALTERNATIVE 1 (Calculation of length)

Read off from graph [$2.8 - 3.2 \times 10^{-5} \Omega \text{ m}$] ✓

Use of $I = \frac{RA}{\rho}$ ✓

$l = 1.3 - 1.4 \text{ «m»}$ ✓

ALTERNATIVE 2 (Calculation of area)

Read off from graph [$2.8 - 3.2 \times 10^{-5} \Omega \text{ m}$] ✓

Use of $A = \rho \frac{I}{R}$ ✓

$A = 8.3 - 9.5 \times 10^{-6} \text{ «m}^2\text{»}$ ✓

ALTERNATIVE 3 (Calculation of resistance)

Read off from graph $[2.8 - 3.2 \times 10^{-5} \Omega \text{ m}]$ ✓

Use of $R = \rho \frac{I}{A}$ ✓

$R = 3.6 - 4.2 \text{ «}\Omega\text{»}$ ✓

ALTERNATIVE 4 (Calculation of resistivity)

Use of $\rho = \frac{RA}{I}$ ✓

$\rho = 3.2 \times 10^{-5} \text{ «}\Omega \text{ m}\text{»}$ ✓

Read off from graph 260 – 280 K ✓

Examiners report

Examiners looked for one of four possible approaches to answering this by calculating an area, length, resistance, or resistivity. Most students attempted one of the last 2. Each required a calculation and a reading taken from the graph. Those that chose the resistance route often found an appropriate resistivity from the graph and used it to calculate the value of resistance required. Those that calculated a resistivity took the data from the question but often neglected the final step of reading a temperature from the graph for the resistivity they had calculated.

(b.ii) The power supply to the pad has a negligible internal resistance.

State and explain the variation in current in the resistor as the temperature of the pad increases.

[2]

Markscheme

«Resistivity and hence» resistance will decrease ✓

«Pd across pad will not change because internal resistance is negligible»
Current will increase ✓

Examiners report

This was generally well answered. Some students did write a description of what happens in terms of electrons in a metal conductor, but most realised the graph was showing something different.

- (c) When there is a current in the resistor, magnetic forces act between the resistor strips.

For the part of the resistor labelled RS,

- (c.i) outline the magnetic force acting on it due to the current in PQ.

[1]

Markscheme

«The force is» away from PQ/repulsive/to the right ✓

Examiners report

This was well answered.

- (c.ii) state and explain the net magnetic force acting on it due to the currents in PQ and TU.

[2]

Markscheme

The magnetic fields «due to currents in PQ and TU» are in opposite directions

OR

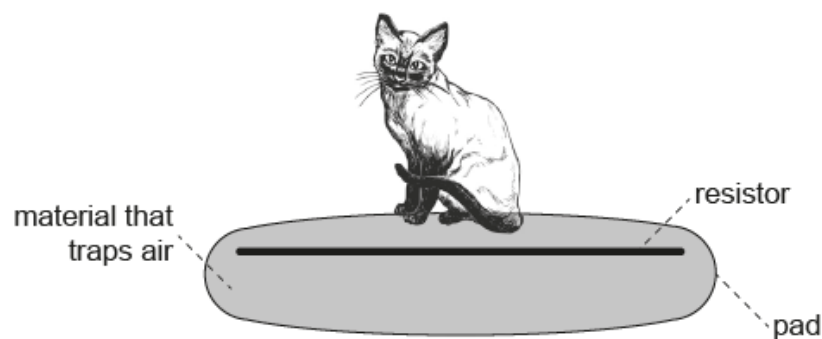
There are two «repulsive» forces in opposite directions ✓

Net force is zero ✓

Examiners report

If students were able to answer the previous part, they were able to answer this one. There were only a few answers that thought the 2 forces added to give a force twice the size.

- (d) The design of the pad encloses the resistor in a material that traps air. The design also places the resistor close to the top surface of the pad.



Explain, with reference to thermal energy transfer, why the pad is designed in this way.

[3]

Markscheme

Air is a poor «thermal» conductor ✓

Lack of convection due to air not being able to move in material ✓

Appropriate statement about energy transfer between the pet, the resistor and surroundings ✓

The rate of thermal energy transfer to the top surface is greater than the bottom «due to thinner material» ✓

Accept air is a good insulator

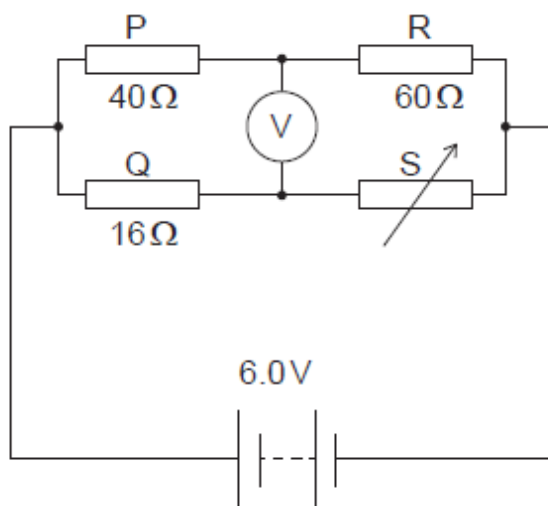
Examiners report

This was one of the lowest scoring questions on the paper. Many students took the description of the material that traps air to mean that most of the thermal energy transfer involved convection currents. There were some very good answers, but these were few and far between.

2. [Maximum mark: 10]

22M.2.SL.TZ1.4

A power supply is connected to three resistors P, Q and R of fixed value and to an ideal voltmeter. A variable resistor S, formed from a solid cylinder of conducting putty, is also connected in the circuit. Conducting putty is a material that can be moulded so that the resistance of S can be changed by re-shaping it.



The resistance values of P, Q and R are $40\ \Omega$, $16\ \Omega$ and $60\ \Omega$ respectively. The emf of the power supply is $6.0\ \text{V}$ and its internal resistance is negligible.

(a) Calculate the potential difference across P.

[2]

Markscheme

ALTERNATIVE 1

attempt to use potential divider equation or similar method ✓

$$\llcorner 6.0 \times \frac{40}{(40+60)} \rrcorner = 2.4 \llcorner \text{V} \rrcorner \checkmark$$

ALTERNATIVE 2

$$\llcorner \text{current} = \frac{6}{60+40} \rrcorner = 0.06 \llcorner \text{A} \rrcorner \checkmark$$

$$40 \times 0.06 = 2.4 \llcorner \text{V} \rrcorner \checkmark$$

Examiners report

This was generally well done at the higher level. Some SL candidates struggled to calculate the correct current but earned the second marking point through ECF.

(b) The voltmeter reads zero. Determine the resistance of S.

[3]

Markscheme

ALTERNATIVE 1

Pd across Q = 2.4 V so $I = 0.15$ « A » ✓

and pd across S is $6.0 - 2.4 = 3.6$ « V » ✓

$$R = \frac{V}{I} = \frac{3.6}{0.15} = 24 \text{ «}\Omega\text{» } \checkmark$$

ALTERNATIVE 2

pd at PR junction = pd at QS junction ✓

$$\text{so } \frac{S}{Q} = \frac{R}{P} \text{ OR } S = \frac{16 \times 60}{40} \checkmark$$

24 « Ω » ✓

ALTERNATIVE 3

$$I = \frac{40 \times 0.06}{16} = 0.15 \text{ «A» } \checkmark$$

$$6.0 = (16 + S)(0.15) \text{ OR } 2.4 = (6) \left(\frac{16}{S+16} \right) \checkmark$$

$$R = 24 \text{ «}\Omega\text{» } \checkmark$$

Allow **ECF** for **MP3** from incorrect **MP1** or **MP2**.

Examiners report

Many candidates struggled with this question. A very common mistake was to assume the current was the same in each branch, leading to a resistance of 84 Ohms. The placement of the voltmeter may have caused some confusion for candidates, and they may not have understood what the zero reading was indicating. It is important that candidates understand what voltmeters are actually reading and are familiar with different placements in circuits.

All the putty is reshaped into a solid cylinder that is four times longer than the original length.

- (c.i) Deduce the resistance of this new cylinder when it has been reshaped.

[3]

Markscheme

recognition that $4L$ leads to $4R / \ll \frac{\rho 4L}{A} \gg = 4R \checkmark$

«because the volume of S is constant new area is» $\frac{A}{4} \checkmark$

$16 \times 24 = 384 \ll \Omega \gg \checkmark$

Examiners report

Most candidates recognized that increasing the length of the conductive putty would increase the resistance by a factor of four, but very few considered that if the volume of the putty remained constant that the cross-sectional surface area would decrease as well.

- (c.ii) Outline, without calculation, the change in the total power dissipated in Q and the new cylinder after it has been reshaped.

[2]

Markscheme

«total» power has decreased ✓

Because current in the branch has decreased «and $P=I^2R$ »

OR

Because resistance has increased in branch «and $P = \frac{V^2}{R}$ » ✓

*Allow opposite argument as **ECF** from (c)(i) (if candidate deduces a lower resistance).*

Allow “power doesn’t change” if candidate has no change of resistance from (b) to (c)(i).

Examiners report

This was an item that caused a bit of confusion for candidates. The prompt asks for a comparison of the power in the branch before and after changing the length of the conductive putty. Many candidates correctly identified that the power in Q would decrease, but either did not discuss the power in the whole branch or were not clear that the power in the putty would decrease as well.

3. [Maximum mark: 14]

22M.2.SL.TZ2.4

- (a) Identify the laws of conservation that are represented by Kirchhoff's circuit laws.

[2]

Markscheme

« conservation of » charge ✓

« conservation of » energy ✓

Allow [1] max if they explicitly refer to Kirchhoff's laws linking them to the conservation laws incorrectly.

Examiners report

a) Most just stated Kirchhoff's laws rather than the underlying laws of conservation of energy and charge, basically describing the equations from the data booklet. When it came to guesses, energy and momentum were often the two, although even a baryon and lepton number conservation was found. It cannot be emphasised enough the importance of correctly identifying the command verb used to introduce the question. In this case, identify, with the specific reference to conservation laws, seem to have been explicit tips not picked up by some candidates.

bi) This was probably the easiest question on the paper and almost everybody got it right. 12V. Some calculations were seen, though, that contradict the command verb used. State a value somehow implies that the value is right in front to be read or interpreted suitably.

bii) In the end a lot of the answers here were correct. Some obtained 2 ohms and were able to provide an explanation that worked. A very few negative answers were found, suggesting that some candidates work mechanically without properly reflecting in the nature of the value obtained.

ci) A lot of candidates figured out they had to do some sort of loop here but most had large currents in the voltmeter. Currents of 2 A and 10 A

simultaneously were common. Some very good and concise work was seen though, leading to correct steps to show a reading of 8V.

cii) This question was cancelled due to an internal reference error. The paper total was adjusted in grade award. This is corrected for publication and future teaching use.

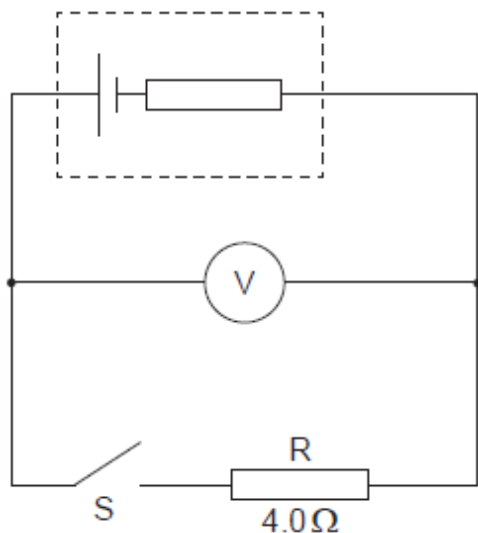
di) The vast majority of candidates could explain why electricity was a secondary energy source.

dii) An ideal answer was that renewable fuels can be replenished faster than they are consumed. However, many imaginative alternatives were accepted.

ei) This question was often very difficult to mark. Working was often scattered all over the answer box. Full marks were not that common, most candidates achieved partial marks. The commonest problem was determining the energy required to charge the battery. It was also common to see a final calculation involving a power divided by a power to calculate the time.

eii) Almost everybody could give a valid reason why research into solar cells was important. Most answers stated that solar is renewable. There were very few that didn't get a mark due to discussing economic reasons.

A cell is connected to an ideal voltmeter, a switch S and a resistor R. The resistance of R is $4.0\ \Omega$.



When S is open the reading on the voltmeter is 12 V. When S is closed the voltmeter reads 8.0 V.

(b.i) State the emf of the cell.

[1]

Markscheme

12 V ✓

(b.ii) Deduce the internal resistance of the cell.

[2]

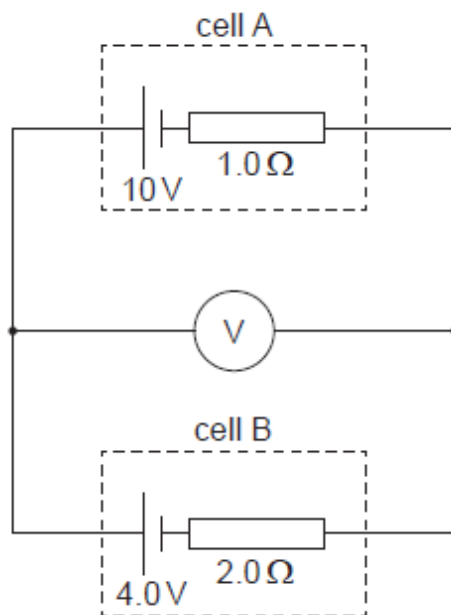
Markscheme

$$I = 2.0 \text{ A} \text{ OR } 12 = I(r + 4) \text{ OR } 4 = Ir \text{ OR } 8 = 4I \text{ ✓}$$

$$\text{«Correct working to get » } r = 2.0 \text{ «}\Omega\text{» ✓}$$

Allow **ECF** from (b)(i)

(c) The voltmeter is used in another circuit that contains two secondary cells.



Cell A has an emf of 10 V and an internal resistance of 1.0 Ω . Cell B has an emf of 4.0 V and an internal resistance of 2.0 Ω .

[3]

Calculate the reading on the voltmeter.

Markscheme

Loop equation showing ***EITHER*** correct voltages, i.e., 10 – 4 on one side or both emfs positive on different sides of the equation ***OR*** correct resistances, i.e. $I(1 + 2)$ ✓

$$10 - 4 = I(1 + 2) \text{ ***OR*** } I = 2.0 \text{ «A» seen ✓}$$

$$V = 8.0 \text{ «V» ✓}$$

Allow any valid method

Electricity can be generated using renewable resources.

(d.i) Outline why electricity is a secondary energy source.

[1]

Markscheme

is generated from primary/other sources ✓

- (d.ii) Some fuel sources are renewable. Outline what is meant by renewable.

[1]

Markscheme

«a fuel » that can be replenished/replaced within a reasonable time span

OR

«a fuel» that can be replaced faster than the rate at which it is consumed

OR

renewables are limitless/never run out

OR

«a fuel» produced from renewable sources

OR

gives an example of a renewable (biofuel, hydrogen, wood, wind, solar, tidal, hydro etc..) ✓

OWTTE

- (e.i) A fully charged cell of emf 6.0 V delivers a constant current of 5.0 A for a time of 0.25 hour until it is completely discharged.

The cell is then re-charged by a rectangular solar panel of dimensions $0.40 \text{ m} \times 0.15 \text{ m}$ at a place where the maximum intensity of sunlight is 380 W m^{-2} .

The overall efficiency of the re-charging process is 18 %.

Calculate the minimum time required to re-charge the cell fully.

[3]

Markscheme

ALTERNATIVE 1

«energy output of the panel $\Rightarrow VIt$ **OR** $6 \times 5 \times 0.25 \times 3600$ **OR** 27000 «J» ✓

«available power $\Rightarrow 380 \times 0.4 \times 0.15 \times 0.18$ **OR** 4.1 «W» ✓

$$t = \frac{27000}{4.1} \Rightarrow 6600 \text{ «s» } \checkmark$$

ALTERNATIVE 2

«energy needed from Sun $\Rightarrow \frac{VIt}{eff}$ **OR** $\frac{6 \times 5 \times 0.25 \times 3600}{0.18}$ **OR** 150000 «J» ✓

« incident power $\Rightarrow 380 \times 0.4 \times 0.15$ **OR** 22.8 «W» ✓

$$t = \frac{150000}{22.8} \Rightarrow 6600 \text{ «s» } \checkmark$$

Allow **ECF** for **MP3**

Accept final answer in minutes (110) or hours (1.8).

(e.ii) Outline why research into solar cell technology is important to society.

[1]

Markscheme

coherent reason ✓

e.g., to improve efficiency, is non-polluting, is renewable, does not produce greenhouse gases, reduce use of fossil fuels

*Do **not** allow economic reasons*

4. [Maximum mark: 5]

21M.2.SL.TZ1.2

A planet is in a circular orbit around a star. The speed of the planet is constant.

- (a.i) Explain why a centripetal force is needed for the planet to be in a circular orbit.

[2]

Markscheme

«circular motion» involves a changing velocity ✓

«Tangential velocity» is «always» perpendicular to centripetal force/acceleration ✓

there must be a force/acceleration towards centre/star ✓

without a centripetal force the planet will move in a straight line ✓

- (a.ii) State the nature of this centripetal force.

[1]

Markscheme

gravitational force/force of gravity ✓

- (b) Determine the gravitational field of the planet.

The following data are given:

Mass of planet = 8.0×10^{24} kg

Radius of the planet = 9.1×10^6 m.

[2]

Markscheme

use of $\frac{GM}{R^2}$ ✓

6.4 «Nkg⁻¹ or ms⁻²» ✓

5. [Maximum mark: 12]

21M.2.SL.TZ1.3

A mass of 1.0 kg of water is brought to its boiling point of 100 °C using an electric heater of power 1.6 kW.

- (a.i) The molar mass of water is 18 g mol⁻¹. Estimate the average speed of the water molecules in the vapor produced. Assume the vapor behaves as an ideal gas.

[2]

Markscheme

$$E_k = \left\langle \frac{3}{2} (1.38 \times 10^{-23}) (373) \right\rangle = 7.7 \times 10^{-21} \text{ «J» } \checkmark$$

$$v = \left\langle \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 373}{0.018}} \right\rangle = 720 \text{ «m s}^{-1}\text{» } \checkmark$$

- (a.ii) State **one** assumption of the kinetic model of an ideal gas.

[1]

Markscheme

particles can be considered points «without dimensions» ✓

no intermolecular forces/no forces between particles «except during collisions» ✓✓

the volume of a particle is negligible compared to volume of gas ✓

collisions between particles are elastic ✓

time between particle collisions are greater than time of collision ✓

no intermolecular PE/no PE between particles ✓

Accept reference to atoms/molecules for "particle"

A mass of 0.86 kg of water remains after it has boiled for 200 s.

- (b.i) Estimate the specific latent heat of vaporization of water. State an appropriate unit for your answer.

[2]

Markscheme

$$\llcorner mL = Pt \gg \text{ so } \llcorner L = \frac{1600 \times 200}{0.14} \gg = 2.3 \times 10^6 \llcorner \text{J kg}^{-1} \gg \checkmark$$

J kg⁻¹ ✓

- (b.ii) Explain why the temperature of water remains at 100 °C during this time.

[1]

Markscheme

«all» of the energy added is used to increase the «intermolecular» potential energy of the particles/break «intermolecular» bonds/**OWTTE** ✓

Accept reference to atoms/molecules for "particle"

- (c) The heater is removed and a mass of 0.30 kg of pasta at –10 °C is added to the boiling water.

Determine the equilibrium temperature of the pasta and water after the pasta is added. Other heat transfers are negligible.

Specific heat capacity of pasta = 1.8 kJ kg⁻¹ K⁻¹

Specific heat capacity of water = 4.2 kJ kg⁻¹ K⁻¹

[3]

Markscheme

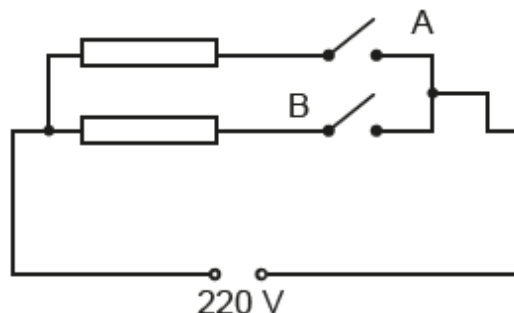
use of $mc\Delta T$ ✓

$$0.86 \times 4200 \times (100 - T) = 0.3 \times 1800 \times (T + 10) \checkmark$$

$$T_{\text{eq}} = 85.69 \llcorner ^\circ\text{C} \gg \cong 86 \llcorner ^\circ\text{C} \gg \checkmark$$

Accept T_{eq} in Kelvin (359 K).

The electric heater has two identical resistors connected in parallel.



The circuit transfers 1.6 kW when switch A only is closed. The external voltage is 220 V.

(d.i) Show that each resistor has a resistance of about 30 Ω .

[1]

Markscheme

$$P = \frac{v^2}{R} \text{ so } \frac{220^2}{1600} \text{ so } R = 30.25 \text{ «}\Omega\text{» } \checkmark$$

Must see either the substituted values **OR** a value for R to at least three s.f.

(d.ii) Calculate the power transferred by the heater when both switches are closed.

[2]

Markscheme

use of parallel resistors addition so $R_{eq} = 15 \text{ «}\Omega\text{» } \checkmark$

$$P = 3200 \text{ «W» } \checkmark$$

6. [Maximum mark: 10]

21M.2.SL.TZ2.6

A photovoltaic cell is supplying energy to an external circuit. The photovoltaic cell can be modelled as a practical electrical cell with internal resistance.

The intensity of solar radiation incident on the photovoltaic cell at a particular time is at a maximum for the place where the cell is positioned.

The following data are available for this particular time:

Operating current = 0.90 A

Output potential difference to external circuit = 14.5 V

Output emf of photovoltaic cell = 21.0 V

Area of panel = 350 mm × 450 mm

- (a) Explain why the output potential difference to the external circuit and the output emf of the photovoltaic cell are different.

[2]

Markscheme

there is a potential difference across the internal resistance

OR

there is energy/power dissipated in the internal resistance ✓

when there is current «in the cell»/as charge flows «through the cell» ✓

Allow full credit for answer based on $V = \mathcal{E} - Ir$

- (b) Calculate the internal resistance of the photovoltaic cell for the maximum intensity condition using the model for the cell.

[3]

Markscheme

ALTERNATIVE 1

pd dropped across cell = 6.5 «V» ✓

internal resistance = $\frac{6.5}{0.9}$ ✓

$7.2 \text{ «}\Omega\text{»}$ ✓

ALTERNATIVE 2

$\mathcal{E} = I(R + r)$ so $\mathcal{E} = V + Ir$ ✓

$21.0 = 14.5 + 0.9 \times r$ ✓

$7.2 \text{ «}\Omega\text{»}$ ✓

Alternative solutions are possible

Award [3] marks for a bald correct answer

- (c) The maximum intensity of sunlight incident on the photovoltaic cell at the place on the Earth's surface is 680 W m^{-2} .

A measure of the efficiency of a photovoltaic cell is the ratio

$$\frac{\text{energy available every second to the external circuit}}{\text{energy arriving every second at the photovoltaic cell surface}} \cdot$$

Determine the efficiency of this photovoltaic cell when the intensity incident upon it is at a maximum.

[3]

Markscheme

power arriving at cell = $680 \times 0.35 \times 0.45 = \text{«107 W»}$ ✓

power in external circuit = $14.5 \times 0.9 = \text{«13.1 W»}$ ✓

efficiency = 0.12 **OR** 12% ✓

Award **[3] marks** for a bald correct answer

Allow **ECF** for **MP3**

- (d) State **two** reasons why future energy demands will be increasingly reliant on sources such as photovoltaic cells.

Reason 1:
.....
Reason 2:
.....

[2]

Markscheme

«energy from Sun/photovoltaic cells» is renewable

OR

non-renewable are running out ✓

non-polluting/clean ✓

no greenhouse gases

OR

does not contribute to global warming/climate change ✓

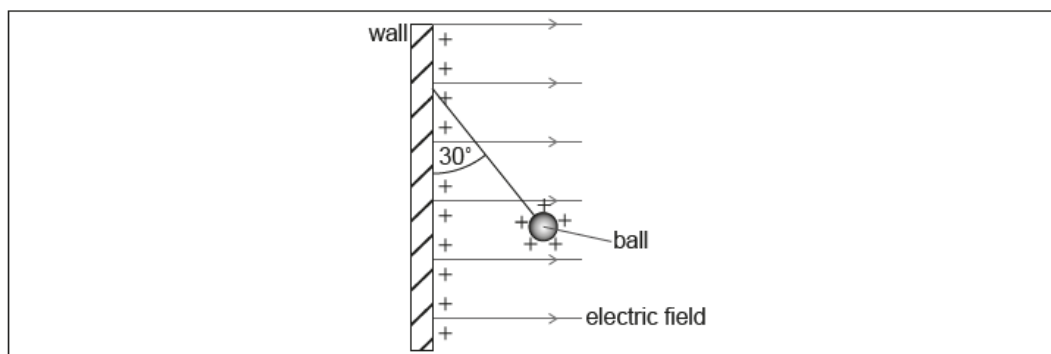
OWTTE

Do not allow economic aspects (e.g. free energy)

7. [Maximum mark: 9]

21M.2.SL.TZ2.3

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



- (a) The charge per unit area on the surface of the wall is σ . It can be shown that the electric field strength E due to the charge on the wall is given by the equation

$$E = \frac{\sigma}{2\epsilon_0}.$$

Demonstrate that the units of the quantities in this equation are consistent.

[2]

Markscheme

identifies units of σ as C m^{-2} ✓

$\frac{\text{C}}{\text{m}^2} \times \frac{\text{Nm}^2}{\text{C}^2}$ seen and reduced to N C^{-1} ✓

Accept any analysis (eg dimensional) that yields answer correctly

- (b.i) The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg.

Determine the horizontal force that acts on the ball.

[3]

Markscheme

horizontal force F on the ball = $T \sin 30$ ✓

$$T = \frac{mg}{\cos 30} \text{ ✓}$$

$$F \ll = mg \tan 30 = 0.025 \times 9.8 \times \tan 30 \gg = 0.14 \ll \text{N} \gg \text{ ✓}$$

Allow $g = 10 \text{ N kg}^{-1}$

Award **[3] marks** for a bald correct answer.

Award **[1 max]** for an answer of zero, interpreting that the horizontal force refers to the horizontal component of the net force.

(b.ii) The charge on the ball is $1.2 \times 10^{-6} \text{ C}$. Determine σ .

[2]

Markscheme

$$E = \frac{0.14}{1.2 \times 10^{-6}} \ll = 1.2 \times 10^5 \gg \text{ ✓}$$

$$\sigma = \ll \frac{2 \times 8.85 \times 10^{-12} \times 0.14}{1.2 \times 10^{-6}} \gg = 2.1 \times 10^{-6} \ll \text{C m}^{-2} \gg \text{ ✓}$$

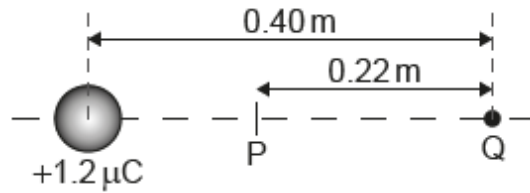
Allow **ECF** from the calculated F in (b)(i)

Award **[2]** for a bald correct answer.

(c) The centre of the ball, still carrying a charge of $1.2 \times 10^{-6} \text{ C}$, is now placed 0.40 m from a point charge Q . The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero.

The distance PQ is 0.22 m.



Calculate the charge on Q. State your answer to an appropriate number of significant figures.

[2]

Markscheme

$$\frac{Q}{0.22^2} = \frac{1.2 \times 10^{-6}}{0.18^2} \checkmark$$

$$\ll + \gg 1.8 \times 10^{-6} \ll \text{C} \gg \checkmark$$

2sf ✓

*Do not award **MP2** if charge is negative*

*Any answer given to 2 sig figs scores **MP3***

8. [Maximum mark: 15]

19M.2.SL.TZ1.1

A girl rides a bicycle that is powered by an electric motor. A battery transfers energy to the electric motor. The emf of the battery is 16 V and it can deliver a charge of 43 kC when discharging completely from a full charge.

The maximum speed of the girl on a horizontal road is 7.0 m s^{-1} with energy from the battery alone. The maximum distance that the girl can travel under these conditions is 20 km.

- (a.i) Show that the time taken for the battery to discharge is about $3 \times 10^3 \text{ s}$.

[1]

Markscheme

$$\text{time taken } \frac{2.0 \times 10^4}{7} \ll 2860 \text{ s} = 2900 \ll \text{s} \checkmark$$

Must see at least two s.f.

Examiners report

This question was generally well answered. Candidates should be reminded on questions where a given value is being calculated that they should include an unrounded answer. This whole question set was a blend of electricity and mechanics concepts, and it was clear that some candidates struggled with applying the correct concepts in the various sub-questions.

- (a.ii) Deduce that the average power output of the battery is about 240 W.

[2]

Markscheme

$$\text{use of } E = qV \text{ OR energy} = 4.3 \times 10^3 \times 16 \ll 6.88 \times 10^5 \text{ J} \checkmark$$

$$\text{power} = 241 \ll \text{W} \checkmark$$

Accept 229 W – 241 W depending on the exact value of t used from ai.

Must see at least three s.f.

- (a.iii) Friction and air resistance act on the bicycle and the girl when they move. Assume that all the energy is transferred from the battery to the electric motor. Determine the total average resistive force that acts on the bicycle and the girl.

[2]

Markscheme

use of power = force $\times$ speed **OR** force $\times$ distance = power $\times$ time ✓

«34N» ✓

Award [2] for a bald correct answer.

Accept 34 N – 36 N.

The bicycle and the girl have a total mass of 66 kg. The girl rides up a slope that is at an angle of 3.0° to the horizontal.



- (b.i) Calculate the component of weight for the bicycle and girl acting down the slope.

[1]

Markscheme

$$66 \text{ g} \sin(3^\circ) = 34 \text{ «N»} \checkmark$$

Examiners report

Many candidates struggled with this question. They either simply calculated the weight, used the cosine rather than the sine function, or failed to multiply by the acceleration due to gravity. Candidates need to be able to apply free-body diagram skills in a variety of “real world” situations.

- (b.ii) The battery continues to give an output power of 240 W. Assume that the resistive forces are the same as in (a)(iii).

Calculate the maximum speed of the bicycle and the girl up the slope.

[2]

Markscheme

total force $34 + 34 = 68 \text{ «N»} \checkmark$

$3.5 \text{ «ms}^{-1}\text{»} \checkmark$

If you suspect that the incorrect reference in this question caused confusion for a particular candidate, please refer the response to the PE.

Look for ECF from aiii and bi.

Accept $3.4 - 3.5 \text{ «ms}^{-1}\text{»}.$

Award [0] for solutions involving use of KE.

Award [0] for $v = 7 \text{ ms}^{-1}.$

Award [2] for a bald correct answer.

- (c) On another journey up the slope, the girl carries an additional mass. Explain whether carrying this mass will change the maximum distance that the bicycle can travel along the slope.

Markscheme

«maximum» distance will decrease **OWTTE ✓**

because opposing/resistive force has increased

OR

because more energy is transferred to GPE

OR

because velocity has decreased

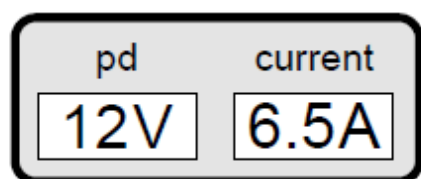
OR

increased mass means more work required «to move up the hill» ✓

Examiners report

This question was well answered in general, with the vast majority of candidates specifying that the maximum distance would decrease. This is an “explain” command term, so the examiners were looking for a detailed reason why the distance would decrease for the second marking point. Unfortunately, some candidates simply wrote that because the mass increased so did the weight without making it clear why this would change the maximum distance.

The bicycle has a meter that displays the current and the terminal potential difference (pd) for the battery when the motor is running. The diagram shows the meter readings at one instant. The emf of the cell is 16 V.



(d) Determine the internal resistance of the battery.

Markscheme

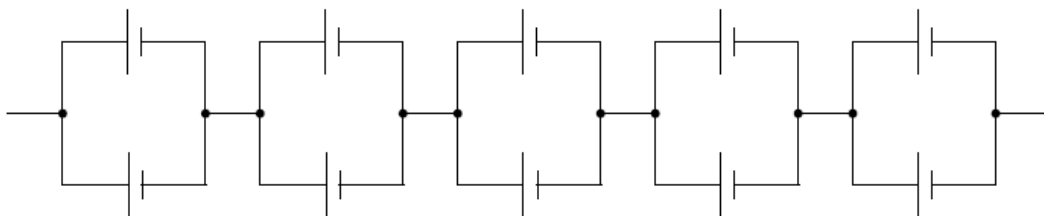
V dropped across battery **OR** $R_{\text{circuit}} = 1.85 \, \Omega$ ✓

so internal resistance = $\frac{4.0}{6.5} = 0.62 \, \Omega$ ✓

For MP1 allow use of internal resistance equations that leads to $16V - 12V (=4V)$.

Award [2] for a bald correct answer.

The battery is made from an arrangement of 10 identical cells as shown.



(e.i) Calculate the emf of **one** cell.

[1]

Markscheme

$$\frac{16}{5} = 3.2 \, \text{V} \, \checkmark$$

(e.ii) Calculate the internal resistance of **one** cell.

[2]

Markscheme

ALTERNATIVE 1:

$$2.5r = 0.62 \, \checkmark$$

$$r = 0.25 \, \Omega \, \checkmark$$

ALTERNATIVE 2:

$$\frac{0.62}{5} = 0.124 \llcorner \Omega \llcorner \checkmark$$

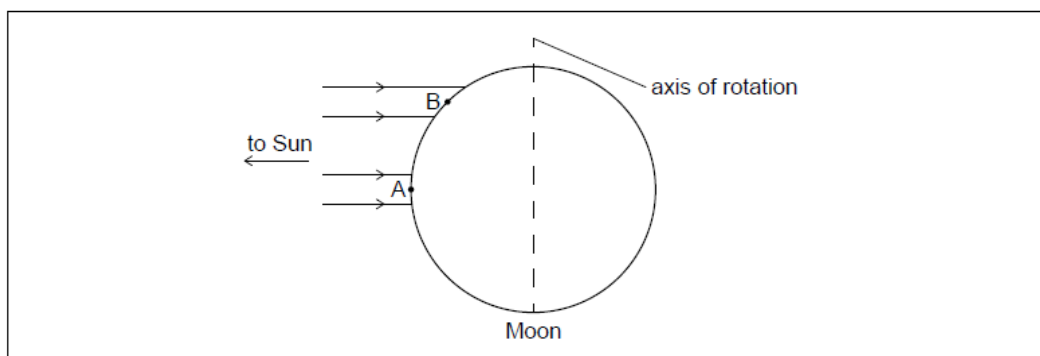
$$r = 2(0.124) = 0.248 \llcorner \Omega \llcorner \checkmark$$

Allow ECF from (d) and/or e(i).

9. [Maximum mark: 7]

19M.2.SL.TZ1.6

The Moon has no atmosphere and orbits the Earth. The diagram shows the Moon with rays of light from the Sun that are incident at 90° to the axis of rotation of the Moon.



- (a.i) A black body is on the Moon's surface at point A. Show that the maximum temperature that this body can reach is 400 K. Assume that the Earth and the Moon are the same distance from the Sun.

[2]

Markscheme

$$T = \left(\frac{1360}{\sigma} \right)^{0.25} \quad \checkmark$$

$$390 \text{ «K»} \quad \checkmark$$

Must see 1360 (from data booklet) used for MP1.

Must see at least 2 s.f.

Examiners report

Many candidates struggled with this question. A significant portion attempted to apply Wein's Law and simply stated that a particular wavelength was the peak and then used that to determine the temperature. Some did use the solar constant from the data booklet and were able to calculate the correct temperature. As part of their preparation for the exam candidates should thoroughly review the data booklet and be aware of

what constants are given there. As with all “show that” questions candidates should be reminded to include an unrounded answer.

- (a.ii) Another black body is on the Moon’s surface at point B.

Outline, without calculation, why the maximum temperature of the black body at point B is less than at point A.

[2]

Markscheme

energy/Power/Intensity lower at B ✓

connection made between energy/power/intensity and temperature of blackbody ✓

Examiners report

This question is another example of candidates not thinking beyond the obvious in the question. Many simply said that point B is farther away, or that it is at an angle. Some used vague terms like “the sunlight is more spread out” rather than using proper physics terms. Few candidates connected the lower intensity at B with the lower temperature of the blackbody.

- (b) The albedo of the Earth’s atmosphere is 0.28. Outline why the maximum temperature of a black body on the Earth when the Sun is overhead is less than that at point A on the Moon.

[1]

Markscheme

(28 %) of sun’s energy is scattered/reflected by earth’s atmosphere **OR** only 72 % of incident energy gets absorbed by blackbody ✓

Must be clear that the energy is being scattered by the atmosphere.

Award [0] for simple definition of “albedo”.

Examiners report

This question was assessing the understanding of the concept of albedo. Many candidates were able to connect that an albedo of 0.28 meant that 28 % of the incident energy from the sun was being reflected or scattered by the atmosphere before reaching the black body.

(c.i) Outline why a force acts on the Moon.

[1]

Markscheme

gravitational attraction/force/field «of the planet/Moon» ✓

Do not accept "gravity".

Examiners report

This was generally well answered, although some candidates simply used the vague term "gravity" rather than specifying that it is a gravitational force or a gravitational field. Candidates need to be reminded about using proper physics terms and not more general, "every day" terms on the exam.

(c.ii) Outline why this force does no work on the Moon.

[1]

Markscheme

the force/field and the velocity/displacement are at 90° to each other **OR** there is no change in GPE of the moon ✓

Award [0] for any mention of no net force on the satellite.

Do not accept acceleration is perpendicular to velocity.

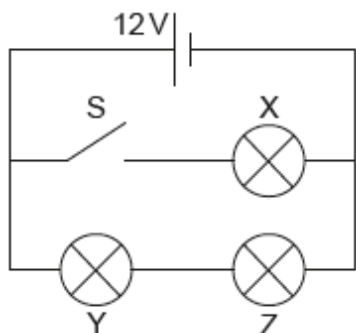
Examiners report

Some candidates connected the idea that the gravitational force is perpendicular to the velocity (and hence the displacement) for the mark. It was also allowed to discuss that there is no change in gravitational potential energy, so therefore no work was being done. It was not acceptable to simply state that the net displacement over one full orbit is zero. Unfortunately, some candidates suggested that there is no net force on the moon so there is no work done, or that the moon is so much smaller so no work could be done on it.

10. [Maximum mark: 5]

19M.2.SL.TZ2.4

Three identical light bulbs, X, Y and Z, each of resistance $4.0\ \Omega$ are connected to a cell of emf 12 V . The cell has negligible internal resistance.



- (a) The switch S is initially open. Calculate the total power dissipated in the circuit.

[2]

Markscheme

total resistance of circuit is $8.0\ \Omega$ ✓

$$P = \frac{12^2}{8.0} = 18\ \text{W} \quad \checkmark$$

Examiners report

Most candidates scored both marks. ECF was awarded for those who didn't calculate the new resistance correctly. Candidates showing clearly that they were attempting to calculate the new total resistance helped examiners to award ECF marks.

- (bi) The switch is now closed. State, without calculation, why the current in the cell will increase.

[1]

Markscheme

«a resistor is now connected in parallel» reducing the total resistance

OR

current through YZ unchanged and additional current flows through X ✓

Examiners report

Most recognised that this decreased the total resistance of the circuit.

Answers scoring via the second alternative were rare as the statements were often far too vague.

(bii) The switch is now closed. Deduce the ratio

$$\frac{\text{power dissipated in Y with S open}}{\text{power dissipated in Y with S closed}} \cdot$$

[2]

Markscheme

evidence in calculation or statement that pd across Y/current in Y is the same as before ✓

so ratio is 1 ✓

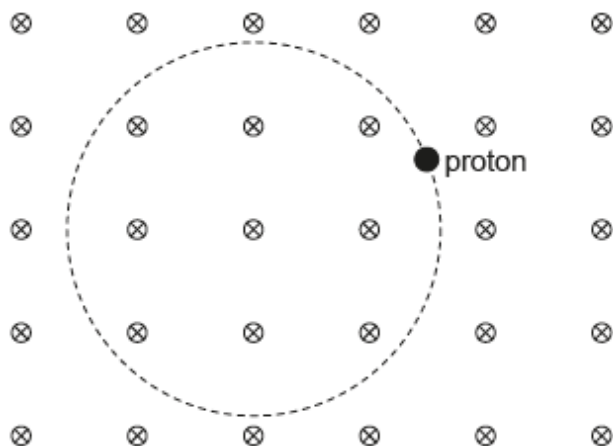
Examiners report

Very few gained any credit for this at both levels. Most performed complicated calculations involving the total circuit and using 12V – they had not realised that the question refers to Y only.

11. [Maximum mark: 5]

19M.2.SL.TZ2.5

A proton moves along a circular path in a region of a uniform magnetic field. The magnetic field is directed into the plane of the page.



- (ai) Label with arrows on the diagram the magnetic force F on the proton.

[1]

Markscheme

F towards centre ✓

Examiners report

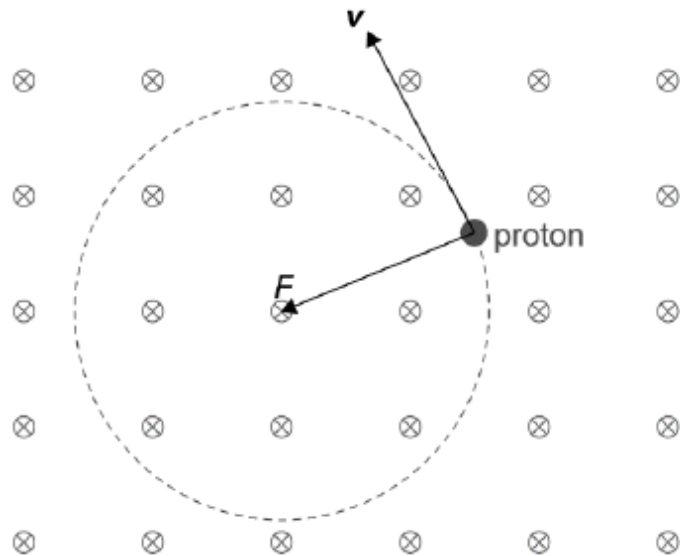
Examiners were requested to be lenient here and as a result most candidates scored both marks. Had we insisted on *e.g.* straight lines drawn with a ruler or a force arrow passing exactly through the centre of the circle very few marks would have been scored. For those who didn't know which way the arrows were supposed to be the common guesses were to the left and up the page. Some candidates neglected to label the arrows.

- (aii) Label with arrows on the velocity vector v of the proton.

[1]

Markscheme

v tangent to circle and in the direction shown in the diagram ✓



Examiners report

Examiners were requested to be lenient here and as a result most candidates scored both marks. Had we insisted on *e.g.* straight lines drawn with a ruler or a force arrow passing exactly through the centre of the circle very few marks would have been scored. For those who didn't know which way the arrows were supposed to be the common guesses were to the left and up the page. Some candidates neglected to label the arrows.

- (b) The speed of the proton is $2.16 \times 10^6 \text{ m s}^{-1}$ and the magnetic field strength is 0.042 T . For this proton, determine, in m , the radius of the circular path. Give your answer to an appropriate number of significant figures.

[3]

Markscheme

$$\llcorner qvB = \frac{mv^2}{R} \Rightarrow \llcorner R = \frac{mv}{qB} / \frac{1.673 \times 10^{-27} \times 2.16 \times 10^6}{1.60 \times 10^{-19} \times 0.042} \quad \checkmark$$

$$R = 0.538 \text{ «m»} \quad \checkmark$$

$$R = 0.54 \text{ «m»} \checkmark$$

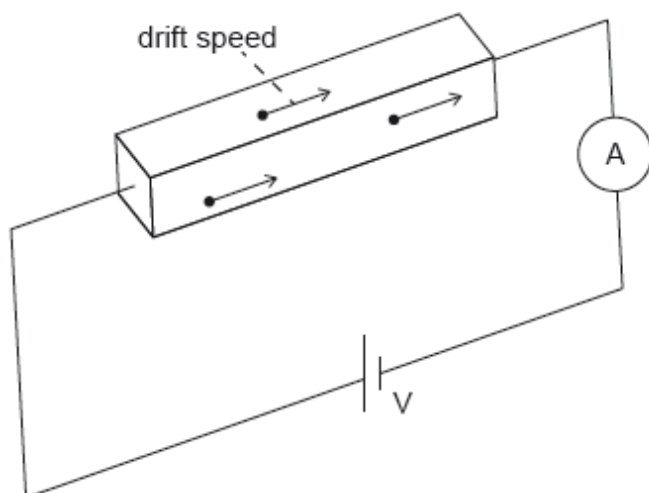
Examiners report

This was generally well answered although usually to 3 sf. Common mistakes were to substitute 0.042 for F and 1 for q. Also some candidates tried to answer in terms of electric fields.

12. [Maximum mark: 5]

18M.2.SL.TZ1.4

An ohmic conductor is connected to an ideal ammeter and to a power supply of output voltage V .



The following data are available for the conductor:

density of free electrons $= 8.5 \times 10^{22} \text{ cm}^{-3}$

resistivity $\rho = 1.7 \times 10^{-8} \Omega\text{m}$

dimensions $w \times h \times l = 0.020 \text{ cm} \times 0.020 \text{ cm} \times 10 \text{ cm}$.

The ammeter reading is 2.0 A.

(a) Calculate the resistance of the conductor.

[2]

Markscheme

$$1.7 \times 10^{-8} \times \frac{0.10}{(0.02 \times 10^{-2})^2}$$

0.043 « Ω »

[2 marks]

- (b) Calculate the drift speed v of the electrons in the conductor in cm s^{-1} . State your answer to an appropriate number of significant figures.

[3]

Markscheme

$$v \llcorner = \frac{I}{neA} \rrcorner = \frac{2}{8.5 \times 10^{22} \times 1.60 \times 10^{-19} \times 0.02^2}$$

$$0.368 \llcorner \text{cm s}^{-1} \rrcorner$$

$$0.37 \llcorner \text{cm s}^{-1} \rrcorner$$

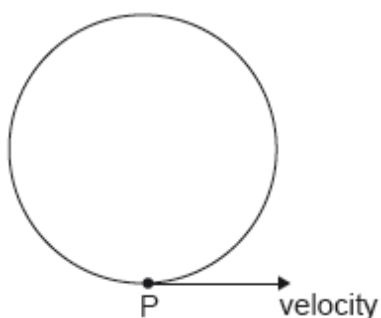
Award [2 max] if answer is not expressed to 2 sf.

[3 marks]

13. [Maximum mark: 5]

18M.2.SL.TZ1.5

An electron moves in circular motion in a uniform magnetic field.



The velocity of the electron at point P is $6.8 \times 10^5 \text{ m s}^{-1}$ in the direction shown.

The magnitude of the magnetic field is 8.5 T.

(a) State the direction of the magnetic field.

[1]

Markscheme

out of the page plane / $\odot$

Do not accept just "up" or "outwards".

[1 mark]

(b) Calculate, in N, the magnitude of the magnetic force acting on the electron.

[1]

Markscheme

$$1.60 \times 10^{-19} \times 6.8 \times 10^5 \times 8.5 = 9.2 \times 10^{-13} \text{ «N»}$$

[1 mark]

(c.i) Explain why the electron moves at constant speed.

[1]

Markscheme

the magnetic force does not do work on the electron hence does not change the electron's kinetic energy

OR

the magnetic force/acceleration is at right angles to velocity

[1 mark]

(c.ii) Explain why the electron moves on a circular path.

[2]

Markscheme

the velocity of the electron is at right angles to the magnetic field

(therefore) there is a centripetal acceleration / force acting on the charge

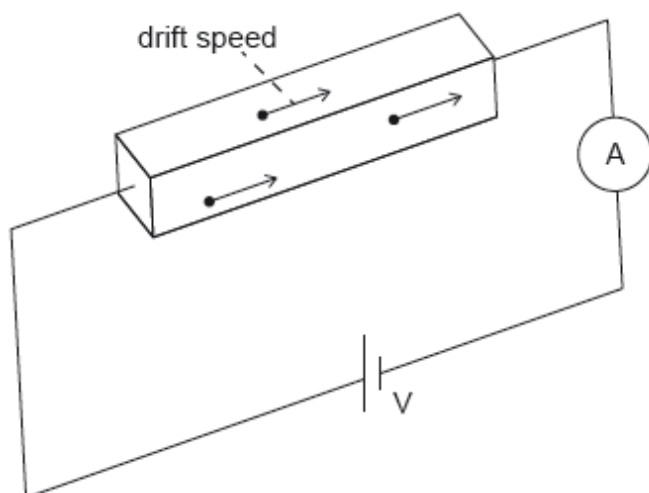
OWTTE

[2 marks]

14. [Maximum mark: 5]

18M.2.SL.TZ1.4

An ohmic conductor is connected to an ideal ammeter and to a power supply of output voltage V .



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resistivity $\rho = 1.7 \times 10^{-8} \Omega\text{m}$

dimensions $w \times h \times l = 0.020 \text{ cm} \times 0.020 \text{ cm} \times 10 \text{ cm}$.

The ammeter reading is 2.0 A.

(a) Calculate the resistance of the conductor.

[2]

Markscheme

$$1.7 \times 10^{-8} \times \frac{0.10}{(0.02 \times 10^{-2})^2}$$

0.043 « Ω »

[2 marks]

- (b) Calculate the drift speed v of the electrons in the conductor in cm s^{-1} . State your answer to an appropriate number of significant figures.

[3]

Markscheme

$$v \llcorner = \frac{I}{neA} \rrcorner = \frac{2}{8.5 \times 10^{22} \times 1.60 \times 10^{-19} \times 0.02^2}$$

$$0.368 \llcorner \text{cm s}^{-1} \rrcorner$$

$$0.37 \llcorner \text{cm s}^{-1} \rrcorner$$

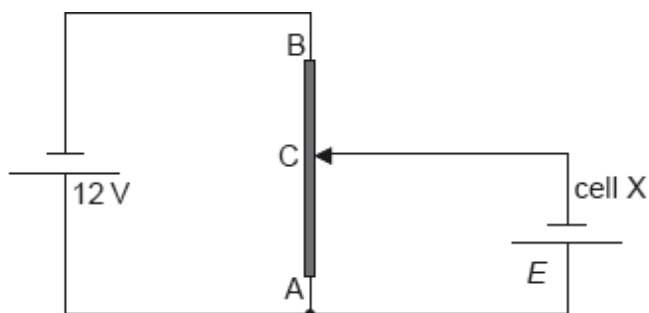
Award [2 max] if answer is not expressed to 2 sf.

[3 marks]

15. [Maximum mark: 6]

18M.2.SL.TZ2.4

The diagram shows a potential divider circuit used to measure the emf E of a cell X. Both cells have negligible internal resistance.



(a) State what is meant by the emf of a cell.

[2]

Markscheme

the work done per unit charge

in moving charge from one terminal of a cell to the other / all the way round the circuit

Award [1] for "energy per unit charge provided by the cell" / "power per unit current"

Award [1] for "potential difference across the terminals of the cell when no current is flowing"

Do not accept "potential difference across terminals of cell"

[2 marks]

AB is a wire of uniform cross-section and length 1.0 m. The resistance of wire AB is $80\ \Omega$. When the length of AC is 0.35 m the current in cell X is zero.

(b.i) Show that the resistance of the wire AC is $28\ \Omega$.

[2]

Markscheme

the resistance is proportional to length / see 0.35 **AND** 1 «.00»

so it equals 0.35×80

«= 28 Ω »

[2 marks]

(b.ii) Determine E .

[2]

Markscheme

current leaving 12V cell is $\frac{12}{80} = 0.15$ «A»

OR

$$E = \frac{12}{80} \times 28$$

$$E = \text{«}0.15 \times 28 = \text{» } 4.2 \text{ «V»}$$

Award [2] for a bald correct answer

Allow a 1sf answer of 4 if it comes from a calculation.

Do not allow a bald answer of 4 «V»

Allow ECF from incorrect current

[2 marks]